

VARYTEC RAZOR DERBY USER MANUAL



Quick Links

[Functions in 2-Channel DMX Mode](#)

[Functions in 9-Channel DMX Mode](#)

Table of Contents

[Table of Contents](#)

[Symbols and signal words](#)

[Switching the device on and off](#)

[Main menu](#)

[Menu overview](#)

[Functions in 2-channel DMX mode](#)

[Functions in 9-channel DMX mode](#)

[Other ManualsLib Projects](#)



Razor Derby

LED Effect

Thomann GmbH
Hans-Thomann-Straße 1
96138 Burgebrach
Germany
Telephone: +49 (0) 9546 9223-0
Internet: www.thomann.de

29.11.2023, ID: 574741

Table of contents

1	General information.....	4
1.1	Symbols and signal words.....	4
2	Safety instructions.....	7
3	Features.....	12
4	Installation.....	13
5	Starting up.....	17
6	Connections and controls.....	19
7	Operating.....	22
7.1	Switching the device on and off.....	22
7.2	Main menu.....	23
7.3	Menu overview.....	30
7.4	Functions in 2-channel DMX mode.....	31
7.5	Functions in 9-channel DMX mode.....	32
8	Technical specifications.....	37
9	Plug and connection assignments.....	40
10	Troubleshooting.....	41
11	Cleaning.....	43
12	Protecting the environment.....	44

1 General information

This document contains important instructions for the safe operation of the product. Read and follow the safety instructions and all other instructions. Keep the document for future reference. Make sure that it is available to all those using the product. If you sell the product to another user, be sure that they also receive this document.

Our products and documentation are subject to a process of continuous development. They are therefore subject to change. Please refer to the latest version of the documentation, which is ready for download under www.thomann.de.

1.1 Symbols and signal words

In this section you will find an overview of the meaning of symbols and signal words that are used in this document.

Signal word	Meaning
DANGER!	This combination of symbol and signal word indicates an immediate dangerous situation that will result in death or serious injury if it is not avoided.
WARNING!	This combination of symbol and signal word indicates a possible dangerous situation that can result in death or serious injury if it is not avoided.
CAUTION!	This combination of symbol and signal word indicates a possible dangerous situation that can result in minor injury if it is not avoided.
NOTICE!	This combination of symbol and signal word indicates a possible dangerous situation that can result in material and environmental damage if it is not avoided.
Warning signs	Type of danger
	Warning – high-voltage.
	Warning – laser radiation.

Warning signs	Type of danger
	Warning – dangerous optical radiation.
	Warning – suspended load.
	Warning – danger zone.

2 Safety instructions

Intended use

This device is intended for use as an electronic lighting effect by means of LED technology and for the projection of laser light effects. The device is designed for professional use only and is not suitable for use in households. Use the device only as described in this user manual. Any other use or use under other operating conditions is considered to be improper and may result in personal injury or property damage. No liability will be assumed for damages resulting from improper use.

This device may be used only by persons with sufficient physical, sensorial, and intellectual abilities and having corresponding knowledge and experience. Other persons may use this device only if they are supervised or instructed by a person who is responsible for their safety.



Extend the operating life of the device by regular breaks and by avoiding frequent switching on and off. The device is not suitable for continuous operation.

Safety



DANGER!

Risk of injury and choking hazard for children!

Children can suffocate on packaging material and small parts. Children can injure themselves when handling the device. Never allow children to play with the packaging material and the device. Always store packaging material out of the reach of babies and small children. Always dispose of packaging material properly when it is not in use. Never allow children to use the device without supervision. Keep small parts away from children and make sure that the device does not shed any small parts (such as knobs) that children could play with.



DANGER!

Danger to life due to electric current!

Within the device there are areas where high voltages may be present. Never remove any covers. There are no user-serviceable parts inside. Do not use the device when covers, safety equipment or optical components are missing or damaged.



DANGER!

Danger to life due to electric current!

A short circuit could lead to a fire hazard and risk of death. Always use proper ready-made insulated triple-core mains cable with a safety plug. Do not modify the mains cable or the plug. In case of isolation damage, disconnect immediately the power supply and arrange repair. If in doubt, seek advice from a qualified electrician.



WARNING!

Risk of eye damage caused by high light intensity!

The device generates highly intense light radiation. Looking directly into the light source can damage the eyes. Never look directly into the light source.

**WARNING!****Risk of epileptic fit due to flashing lights!**

The device emits flashing lights (strobe effects). Flashing lights can trigger epileptic fits in specific people. If you are at risk of epilepsy, avoid spending longer periods of time subjected to flashing lights and looking into strobing light.

**CAUTION!****Risk of injury from laser!**

This device contains a class 2M laser, classified according to EN 60825-1:2014+A11:2021. The accessible laser radiation is in a wavelength range of 400 nm to 700 nm, the output is 1 mW or less. If the beam cross-section is bundled by optical collecting instruments, looking into the laser beam can cause injuries to the eyes and affect the person's vision temporarily. As the operator, observe the applicable legal duties regarding safe operation before putting the device into service. Install the laser system in such a way that persons are not at risk while it is in operation. Do not look into the laser beam for any longer period of time. Do not use optical collecting instruments (such as magnifying glasses or lenses) to look into the laser beam, and ensure that laser beams are not directed to areas where optical collecting instruments are used. Have all work and repairs on class 2M lasers performed by trained experts. Never open a class 2M laser and do not make any technical changes. If you are unsure how to handle the laser safely, include a laser protection representative in your planning.

**CAUTION!****Risk of injury due to stray laser radiation!**

Additional components inadequately secured to the device as well as reflective objects and surfaces at the operating location can cause stray laser radiation, which can cause injuries. Make sure that additional components are always secured adequately. Make sure there are no reflective objects or surfaces in the range of the laser beams.

**NOTICE!****Risk of fire due to covered vents and neighbouring heat sources!**

If the vents of the device are covered or the device is operated in the immediate vicinity of other heat sources, the device can overheat and burst into flames. Never cover the device or the vents. Do not install the device in the immediate vicinity of other heat sources. Never operate the device in the immediate vicinity of naked flames.



NOTICE!

Damage to the device if operated in unsuitable ambient conditions!

The device can be damaged if it is operated in unsuitable ambient conditions. Only operate the device indoors within the ambient conditions specified in the "Technical specifications" chapter of this user manual. Avoid operating it in environments with direct sunlight, heavy dirt and strong vibrations. Avoid operating it in environments with strong temperature fluctuations. If temperature fluctuations cannot be avoided (for example after transport in low outside temperatures), do not switch on the device immediately. Never subject the device to liquids or moisture. Never move the device to another location while it is in operation. In environments with increased dirt levels (for example due to dust, smoke, nicotine or mist): Have the device cleaned by qualified specialists at regular intervals to prevent damage due to overheating and other malfunctions.



NOTICE!

Damage to the device due to high voltages!

The device can be damaged if it is operated with the incorrect voltage or if high voltage peaks occur. In the worst case, excess voltages can also cause a risk of injury and fires. Make sure that the voltage specification on the device matches the local power grid before plugging in the device. Only operate the device from professionally installed mains sockets that are protected by a residual current circuit breaker (FI). As a precaution, disconnect the device from the power grid when storms are approaching or if the device will not be used for a longer period.



NOTICE!

Risk of fire due to installation of a wrong fuse!

Using fuses of a different type than compatible with the device may cause a fire and seriously damage the device. Only use fuses of the same type. Observe the labelling on the device casing and the information in the "Technical data" chapter.



NOTICE!

Risk of fire due to incorrect polarity!

Incorrectly inserted batteries may cause fires and destroy the device and the batteries. Observe the markings on the batteries and on the device. Ensure that proper polarity is observed when inserting batteries.



NOTICE!

Possible damage due to leaking batteries!

Batteries can leak and cause permanent damage to the device. Take the batteries out of the device if it is not going to be used for an extended period of time.



NOTICE!

Possible staining due to plasticiser in rubber feet!

The plasticiser contained in the rubber feet of this product may react with the coating of the floor and cause permanent dark stains after some time. If necessary, use a suitable mat or felt slide to prevent direct contact between the device's rubber feet and the floor.

Duties of the operator

As the operator of the laser system, you must comply with legal occupational safety obligations as per the OStrV (in Germany). The following applies in particular:

- Before putting into service, draw up an assessment of the risk posed by the direct and indirect effects of laser beams in line with the regulations applicable at the operating location.
- Apply appropriate technical safety measures (e.g. shielding, protective walls, barriers) on the basis of the exposure limits determined in the risk assessment.
- Restrict the range of the laser beams to a limited area in order to reduce the risk to persons as much as possible.
- Use technical and organisational safety measures to restrict access to the running laser system for unauthorised persons.
- Instruct staff on the basis of the risk assessment and familiarise them with the necessary safety measures.

3 Features

The device features a unique combination of laser and LED effects.

Special features of the device:

- Laser configuration:
 - Red (650 nm), 100 mW
 - Green (532 nm), 30 mW
- LED configuration:
 - Derby effect: 4 LEDs (1 × red, 1 × green, 1 × blue, 1 × white, 30 W each)
 - Strobe effect: 16 × RGB LEDs, 0.5 W each
- Control via DMX (2 or 9 channels, depending on operation mode), buttons and display on the device as well as the supplied remote control
- Pre-programmed automatic shows
- Sound control
- Stand-alone mode

For technological reasons, the light output of LEDs decreases over their lifetime. This effect increases with higher operating temperature. You can extend the service life of the illuminants by providing adequate ventilation and operating the LEDs with the lowest possible brightness.

4 Installation

Unpack and check carefully there is no transportation damage before using the unit. Keep the equipment packaging. To fully protect the product against vibration, dust and moisture during transportation or storage use the original packaging or your own packaging material suitable for transport or storage, respectively.

You can install the device standing or hanging. When in use, the device must be mounted at a solid surface or clamped to an approved truss.

Work from a stable platform whenever you install or move the device or when you perform any kind of maintenance. Block access under the work area.



WARNING!

Risk of injury from falling devices that were inadequately secured!

If devices are not properly secured during assembly, they can cause severe injury and considerable damage by falling.

When installing and operating, make sure to follow the standards and regulations that apply in your country.

Always secure the device with a secondary safety attachment, such as a safety cable or a safety chain.



CAUTION!

Risk of injury due to stray laser radiation!

Additional components inadequately secured to the device as well as reflective objects and surfaces at the operating location can cause stray laser radiation, which can cause injuries.

Make sure that additional components are always secured adequately.

Make sure there are no reflective objects or surfaces in the range of the laser beams.



CAUTION!

Risk of injury due to improper installation!

There is a risk of injury if a laser is set up incorrectly and the installation point is too low or too close to persons.

In order to comply with the maximum legal exposure limits, determine the actual radiation intensities at the operating location as well as the distances for laser attachment, and install the laser in line with the ambient conditions.

Make sure the laser is firmly attached, and prevent the laser beam from leaving the planned laser beam area (e.g. through screens, housings or software-specific direction restrictions).

Shield laser beams from each other if multiple lasers are operated simultaneously.

Install the laser in such a way that the laser beam does not enter any traffic zones.



NOTICE!

Risk of overheating and fire due to inadequate distance and bad ventilation!

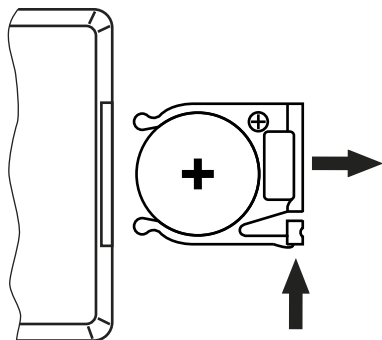
If the distance between the light source and the illuminated surface is too short or the device is badly ventilated, the device can overheat and cause fires.

Make sure that illuminated surfaces are more than 2 m away.

Do not operate the device in ambient temperatures above 40 °C.

Always ensure sufficient ventilation at the operating location.

Inserting the battery into the remote control



Push the lock of the battery holder towards the centre of the housing and pull out the battery holder like a drawer. Insert the batteries. The battery is correct if the positive pole points to the housing base of the remote control. Slide the battery holder back into the remote until it clicks into place.

When shipping, the battery is already installed in the remote and protected against discharge by a transparent plastic film. Remove the plastic film before initial use.



NOTICE!

Risk of fire due to incorrect polarity!

Incorrectly inserted batteries may cause fires and destroy the device and the batteries.

Observe the markings on the batteries and on the device.

Ensure that proper polarity is observed when inserting batteries.



NOTICE!

Possible damage due to leaking batteries!

Batteries can leak and cause permanent damage to the device.

Take the batteries out of the device if it is not going to be used for an extended period of time.

5 Starting up

Create all connections while the device is off. Use the shortest possible high-quality cables for all connections. Take care when running the cables to prevent tripping hazards.



NOTICE!

Data transfer errors due to improper wiring!

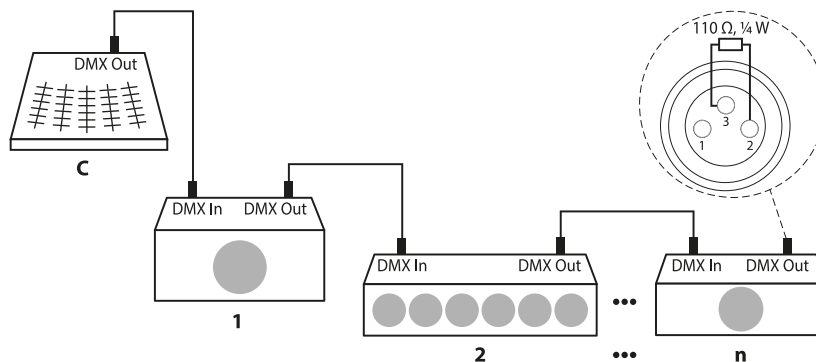
If the DMX connections are wired incorrectly, this can cause errors during the data transfer.

Do not connect the DMX input and output to audio devices, e.g. mixers or amplifiers.

Use special DMX cables for the wiring instead of normal microphone cables.

Connections in DMX mode

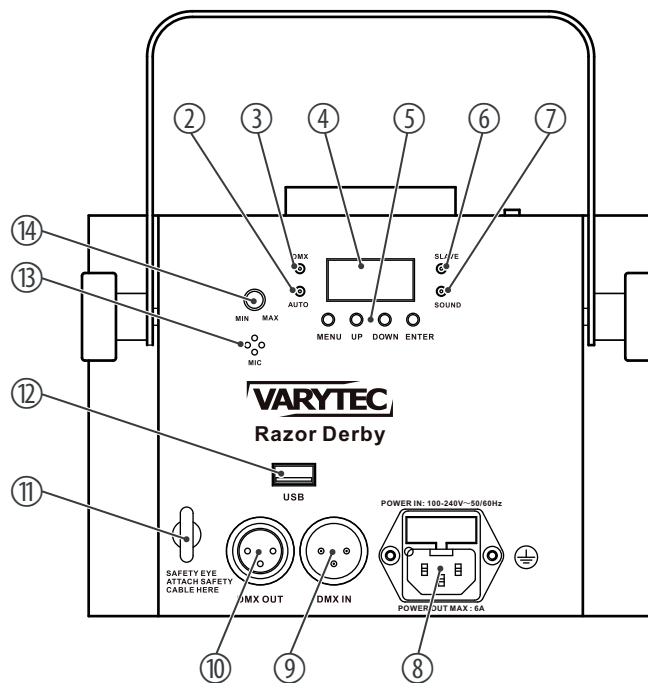
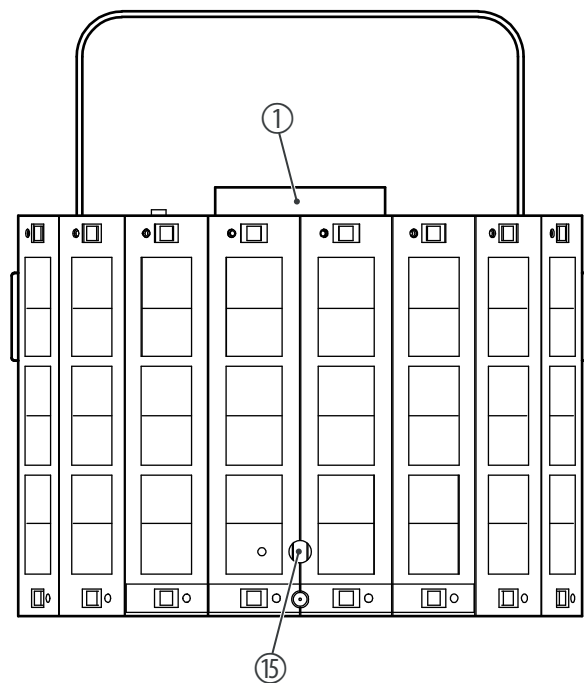
Connect the DMX input of the device to the DMX output of a DMX controller or another DMX device. Connect the output of the first DMX device to the input of the second one, and so on to form a daisy chain. Always ensure that the output of the last DMX device in the daisy chain is terminated with a resistor ($110\ \Omega$, $\frac{1}{4}\text{ W}$).



Connections in master/slave mode

When you configure a group of devices in master/slave mode, the first unit will control the other units for an automatic, sound-activated, synchronized show. This function is ideal when you want to start a show immediately. Connect the DMX output of the master device to the DMX input of the first slave device. Then connect the DMX output of the first slave device to the DMX input of the second slave device and so on.

6 Connections and controls

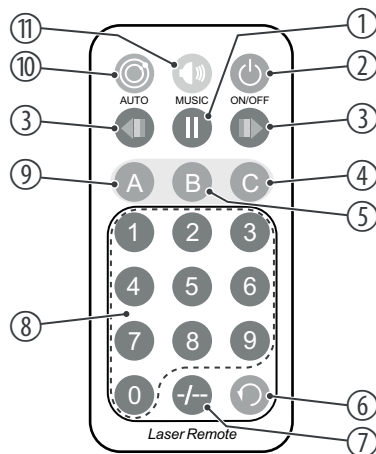


1	Ventilation
2	[<i>AUTO</i>] The LED lights up blue when an automatic show is running.
3	[<i>DMX</i>] The LED lights up red if DMX is connected.
4	LED display
5	[<i>MENU</i>] Activates the main menu and switches between menu items. Closes an open submenu.
	[<i>UP</i>] Increases the displayed value by one.
	[<i>DOWN</i>] Decreases the displayed value by one.
	[<i>ENTER</i>] Selects an option of the respective operating mode, confirms the set value.
6	[<i>SLAVE</i>] The LED lights up yellow if the device is in “slave” mode.
7	[<i>SOUND</i>] The LED lights up green when sound control is active.
8	[<i>POWER</i>] IEC chassis plug with fuse holder for power supply
9	[<i>DMX IN</i>] DMX input
10	[<i>DMX OUT</i>] DMX output
11	Safety cable eyelet
12	USB port for firmware updates
13	[<i>MIC</i>] Microphone for sound control

14 [MIN/MAX] | Control for the sensitivity of the built-in microphone

15 Laser aperture

IR remote control



1 Stops or restarts a running show.

2 [ON/OFF] | Turns the device on and off.

3 Colour change

4 [C] | Turns the strobe effect on and off.

5 [B] | Strobe effect selection.

6 No function

7 No function

8 Numeric buttons for setting the running speed (0: slow, 9: fast)

9 [A] | Turns the derby effect of the four-colour LEDs on and off

10 [AUTO] | Starts an automatic show in random order.

11 [MUSIC] | Starts a sound-controlled automatic show in random order.

7 Operating

7.1 Switching the device on and off

Switching on

Perform the following steps to switch the device on:

1. ► Verify that all required laser safety precautions have been taken. Make sure that there is no one in the reach of the laser beam.
2. ► Connect an external safety switch (e.g. emergency stop switch) or an equivalent system with a protection function to the *[REMOTE]* connection.
3. ► Insert the safety key into the lock.
4. ► If not already done, connect the device to the mains. After a few seconds, the fan and the motors start to work. The display shows the current operation mode. The device is now operational.
5. ► Turn the safety key to the “ON” position to turn the laser beam on.

Turning off

Perform the following steps to switch the device off:

1. ► Turn the safety key to the “OFF” position to turn the laser beam off and remove the key. Keep the safety key in a secure place.
2. ► Disconnect the device from the mains.

7.2 Main menu

Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to select one of the following menu items: 'Aut', 'Sou', 'CoL', 'dNH', 'SLA' or 'SYS'. Press *[ENTER]* to open the selected menu.

Change the currently displayed value with *[UP]* and *[DOWN]* and confirm all new values with *[ENTER]*. To return to the main menu without making changes, press *[MENU]*.

All settings are saved, even if you disconnect the device from the power supply.

Auto menu

In this menu, you can select an automatic show that optionally controls the derby effect LEDs, the laser, the strobe effect or a combination thereof.

Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to select the 'Aut' menu item and confirm with *[ENTER]*. Use *[UP]* and *[DOWN]* to select a show. To confirm and to set the programme speed, press *[ENTER]*. Use *[UP]* and *[DOWN]* to select a value between 'S.01' (slow) and 'S.99' (fast). Confirm with *[ENTER]*.

Show group	Elements used
'Aut1'	Derby effect LEDs + laser + strobe effect
'Aut2'	Derby effect LEDs + strobe effect
'Aut3'	Derby effect LEDs + laser
'Aut4'	Strobe effect + laser
'Aut5'	Derby effect LEDs
'Aut6'	Strobe effect
'Aut7'	Laser

Sound menu

In this menu, you can select an automatic sound-controlled show that optionally controls the derby effect LEDs, the laser, the strobe effect or a combination thereof.

Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to select the 'Sou' menu item and confirm with *[ENTER]*. Use *[UP]* and *[DOWN]* to select a show and confirm with *[ENTER]*.

You can use the *[SENSITIVITY MIN/MAX]* control to adjust the sensitivity of the built-in microphone.

Show group	Elements used
'Sou1'	Derby effect LEDs + laser + strobe effect
'Sou2'	Derby effect LEDs + strobe effect
'Sou3'	Derby effect LEDs + laser
'Sou4'	Strobe effect + laser
'Sou5'	Derby effect LEDs
'Sou6'	Strobe effect
'Sou7'	Laser

Colour menu

In this menu you can adjust the colour of the LEDs for the derby effect. In addition, a strobe effect can be activated for these LEDs.

Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to select the 'CoL' menu item and confirm with *[ENTER]*.

Change the currently displayed parameter with *[UP]* and *[DOWN]* and confirm with *[ENTER]*. To return to the main menu without making changes, press *[MENU]*. To adjust the speed of the strobe effect, press *[ENTER]*. Use *[UP]* and *[DOWN]* to select a value between 'F.00' (no strobe effect) and 'F.99' (fast).

Parameter	Meaning
'r000' ... 'r255'	Red intensity
'G000' ... 'G255'	Green intensity
'b000' ... 'b255'	Blue intensity
'U000' ... 'U255'	White intensity

DMX operating mode

In this mode the device is controlled via a DMX controller.

Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to select the 'dNH' menu item and confirm with *[ENTER]*.

The display shows 'd001'; press *[ENTER]*. Use *[UP]* and *[DOWN]* to select a DMX mode: 2-channel ('02Ch') or 9-channel ('09Ch') and confirm with *[ENTER]*. Now you can set the number of the first DMX channel to be used by the device (DMX address). Use *[UP]* and *[DOWN]* to select a value between 001 and 511 (display shows '001' ... '511'). Confirm with *[ENTER]*.

Make sure that this number matches the configuration of your DMX controller. The following table shows the highest possible DMX address for the various DMX modes.

Mode	Highest possible DMX address
2-channel	511
9-channel	504

To return to the main menu without making changes, press *[MENU]*.

SLAVE mode

In this mode, the device exactly follows the operation of the master that it is connected to.

Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to open the 'SLA' menu and confirm with *[ENTER]*.

The display shows 'SLAv', the device is now in slave mode. If the display is flashing, no master is connected.

To return to the main menu without making changes, or to change back to an active operating mode, press *[MENU]*.

System menu

In this menu, you can adjust various device settings.

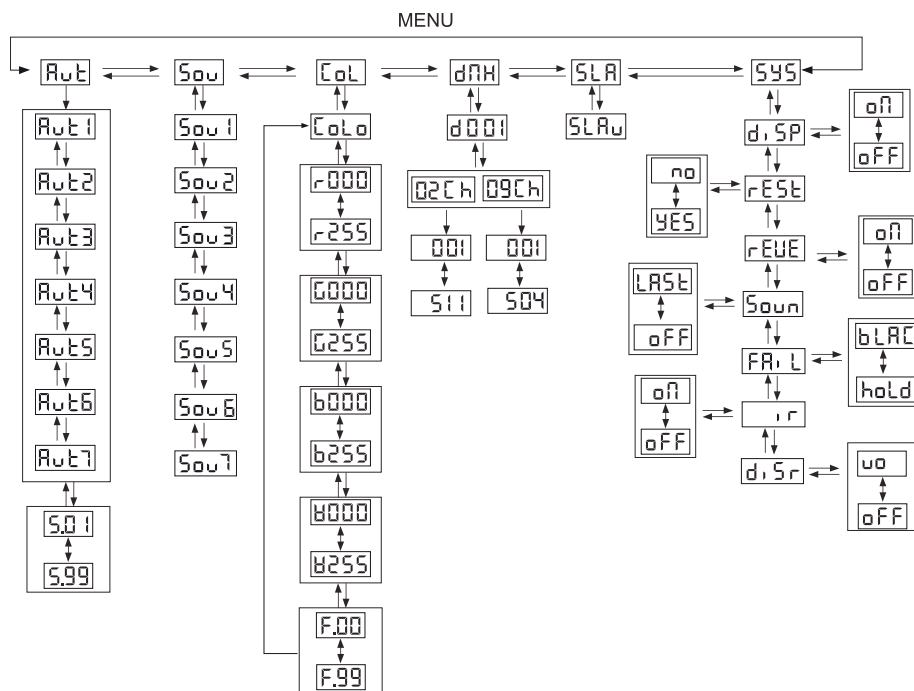
Press *[MENU]* to open the main menu.

Press *[MENU]* repeatedly or press *[UP]* and *[DOWN]* to select the 'SYS' menu and confirm with *[ENTER]*.

Change the currently displayed parameter with *[UP]* and *[DOWN]* and confirm with *[ENTER]*. To return to the main menu without making changes, press *[MENU]*.

Parameter	Meaning	Value range
'diSP'	Automatic display blackout	'oN', 'oFF'
'rEst'	Reset to factory defaults	'no', 'YES'
'rEUE'	Display rotation by 180 °	'oN', 'oFF'
'Soun'	Behaviour if no sound signal is received during a sound-controlled automatic show	'LAST' (sound control remains active), 'oFF' (sound control is turned off)
'FAiL'	Behaviour in the event of DMX control failure	'bLAC' (spotlight is turned off), 'hoLd' (most recent DMX value is maintained)
'Ir'	Sensor for the IR remote control	'oN', 'oFF'
'diSr'	Display shuts off after 5 s	'uo', 'oFF'

7.3 Menu overview



7.4 Functions in 2-channel DMX mode

Channel	Value	Function
1	Selecting a pre-programmed automatic show	
	000...009	Blackout
	010...044	Derby effect LEDs + laser + strobe effect
	045...079	Derby effect LEDs + strobe effect
	080...114	Derby effect LEDs + laser
	115...149	Laser + strobe effect
	150...184	Derby effect LEDs
	185...219	Laser
	220...255	Strobe effect
2	000...250	Progress speed of the selected show, slow to fast
	251...255	Sound control for the selected show

7.5 Functions in 9-channel DMX mode

Channel	Value	Function
1	Selecting an operating mode or a pre-programmed automatic show	
	000...009	Blackout
	010...044	Derby effect LEDs + laser + strobe effect
	045...079	Derby effect LEDs + strobe effect
	080...114	Derby effect LEDs + laser
	115...149	Laser + strobe effect
	150...184	Derby effect LEDs
	185...219	Laser
	220...255	Strobe effect
2	000...250	Progress speed of the selected show, slow to fast
	251...255	Sound control for the selected show
3	Derby effect, if channel 1 = 000...009	
	000...005	No function
	006...020	Red

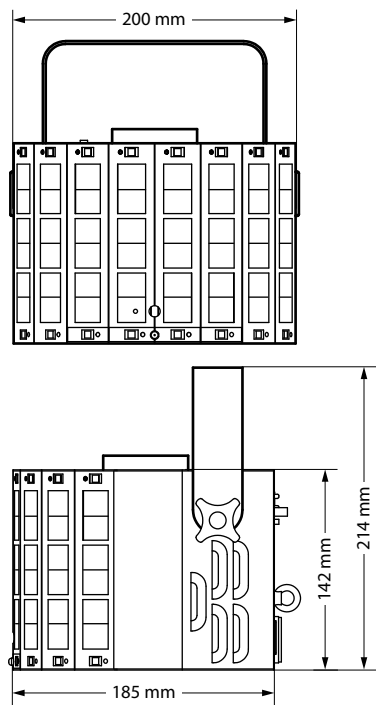
Channel	Value	Function
	021...035	Green
	036...050	Blue
	051...065	White
	066...080	Red + green
	081...095	Red + blue
	096...110	Red + white
	111...125	Green + blue
	126...140	Green + white
	141...155	Blue + white
	156...170	Red + green + blue
	171...185	Red + green + white
	186...200	Green + blue + white
	201...215	Red + green + blue + white
	216...230	Automatic colour change (4 colours)
	231...255	Automatic colour change (7 colours)
4	Strobe effect (derby effect LEDs)	

Channel	Value	Function
	000...0005	Strobe effect off
	006...255	Strobe effect, increasing speed
5	Motor (derby effect LEDs)	
	000	No function
	001...127	Fixed, adjustable position
	128...255	Rotation speed
6	Pattern	
	000...009	Blackout
	010...019	Pattern 1, increasing speed
	020...029	Pattern 2, increasing speed
	030...039	Pattern 3, increasing speed
	040...049	Pattern 4, increasing speed
	050...059	Pattern 5, increasing speed
	060...069	Pattern 6, increasing speed
	070...079	Pattern 7, increasing speed
	080...089	Pattern 8, increasing speed

Channel	Value	Function
	090...099	Pattern 9, increasing speed
	100...109	Pattern 10, increasing speed
	110...119	Pattern 11, increasing speed
	120...129	Pattern 12, increasing speed
	130...139	Pattern 13, increasing speed
	140...149	Pattern 14, increasing speed
	150...159	Pattern 15, increasing speed
	160...169	Pattern 16, increasing speed
	170...179	Pattern 17, increasing speed
	180...255	Pattern 18, increasing speed
7	Laser	
	000...009	Laser off, blackout
	010...049	Red
	050...089	Green
	090...129	Red + green
	130...169	Red + green strobe effect

Channel	Value	Function
	170...209	Green + red strobe effect
	210...255	Red + green (optional strobe effect)
8	Strobe effect (laser)	
	000...009	No function
	010...254	Strobe effect, increasing speed
	255	Sound-controlled strobe effect
9	Rotation of laser projection axis	
	000...004	No rotation
	005...127	Clockwise rotation, speed increasing
	128...133	No rotation
	134...255	Anti-clockwise rotation, speed increasing

8 Technical specifications



Laser medium	Red: 650 nm	
	Green: 532 nm	
Laser power	Red: 100 mW	
	Green: 30 mW	
Laser class	2M	
Light source	Derby effect: 4 LEDs (1 × red, 1 × green, 1 × blue, 1 × white), 30 W each	
	16 × RGB strobe LEDs, 0.5 W each	
Control	DMX	
	Buttons and display on the device	
	IR remote control	
Number of DMX channels	2 or 9	
Input connections	Power supply	IEC chassis plug C14
	DMX control	XLR chassis plug, 3-pin
Output connections	DMX control	XLR chassis socket, 3-pin

Power consumption	46 W	
Supply voltage	100 - 240 V ~ 50/60 Hz	
Fuse	5 mm × 20 mm, 1 A, 250 V, slow blow	
Battery remote control	Battery type	Lithium-ions button cell CR2025
	Voltage	3 V
International Protection Rating	IP20	
Mounting options	Hanging, standing	
Dimensions without bracket (W × H × D)	200 mm × 142 mm × 185 mm	
Dimensions with bracket (W × H × D)	200 mm × 214 mm × 185 mm	
Weight	1.95 kg	
Ambient conditions	Temperature range	0 °C...40 °C
	Relative humidity	20%...80% (non-condensing)

Further information

Similar design	Combination device
DMX control	yes
Master/Slave	yes
Remote control	included
Sound control	yes
Display	yes

9 Plug and connection assignments

Introduction

This chapter will help you select the right cables and plugs to connect your valuable equipment so that a perfect light experience is guaranteed.

Please take our tips, because especially in 'Sound & Light' caution is indicated: Even if a plug fits into a socket, the result of an incorrect connection may be a destroyed DMX controller, a short circuit or 'just' a not working light show!

DMX connections



The unit offers a 3-pin XLR socket for DMX output and a 3-pin XLR plug for DMX input. Please refer to the drawing and table below for the pin assignment of a suitable XLR plug.

Pin	Configuration
1	Ground, shielding
2	Signal inverted (DMX–, 'cold signal')
3	Signal (DMX+, 'hot signal')

10 Troubleshooting

**CAUTION!****Risk of injury due to improper troubleshooting!**

There is a risk of injury if troubleshooting is not performed properly.

Have all work and repairs on class 2M lasers performed by trained experts.

In the following we list a few common problems that may occur during operation. We give you some suggestions for easy troubleshooting:

Symptom	Remedy
The device does not work, no light	1. Check the mains connection and the fuse.
	2. Check the safety key switch.
	3. Check the external safety switch (e.g. emergency stop button)
No response to the DMX controller	1. If the display shows a flashing number, for example "001", no DMX signal is being received. Check the DMX connectors and cables for proper connection.
	2. Check whether the DMX cables run near or parallel to high-voltage cables that may cause damage or interference to a DMX interface circuit.
	3. Try using another DMX controller.
No response to the remote control	1. Try using the remote control at a different angle to the receiver on the front of the device.
	2. Check the remote control battery.

If the procedures recommended above do not succeed, please contact our Service Center. You can find the contact information at www.thomann.de.

11 Cleaning

**CAUTION!****Risk of injury due to improper cleaning!**

There is a risk of injury if cleaning is not performed properly.

To avoid unintentional laser radiation, switch off the laser before you start cleaning the device.

Optical lenses

Clean the optical lenses, that are accessible from the outside, regularly in order to optimize the light output. The frequency of cleaning depends on the operating environment: wet, smoky or particularly dirty surroundings can cause more accumulation of dirt on the optics of the device.

- Clean with a soft cloth using our lamp and lens cleaner (item no. 280122).
- Always dry the parts carefully.

Fan grids

The fan grids of the device must be cleaned of any contamination, such as dust, etc. on a regular basis. Before cleaning, switch off the device and disconnect mains-operated devices from the mains. Only use pH-neutral, solvent-free and non-abrasive cleaning agents. Clean the unit with a slightly damp lint-free cloth.

12 Protecting the environment

Disposal of the packing material



Environmentally friendly materials have been chosen for the packaging. These materials can be sent for normal recycling. Ensure that plastic bags, packaging, etc. are disposed of in the proper manner.

Do not dispose of these materials with your normal household waste, but make sure that they are collected for recycling. Please follow the instructions and markings on the packaging.



Observe the disposal note regarding documentation in France.

Disposal of batteries



Batteries do contain some hazardous chemicals so they should not be thrown away with the normal household waste. They should be returned to the manufacturer for disposal or recycled elsewhere in accordance with your local regulations.

Only dispose of lithium batteries when they are empty. Remove lithium batteries from the device before disposal if this is possible without destroying it. Protect used lithium batteries against short circuit, for example by taping the poles. Dispose the built-in lithium batteries together with the device. Check for an appropriate collection facility.

Dispose of the batteries and rechargeable batteries at relevant collection points or through your local waste facility.

Disposal of your old device



This product is subject to the European Waste Electrical and Electronic Equipment Directive (WEEE) as amended.

Do not dispose of your old device with your normal household waste; instead, deliver it for controlled disposal by an approved waste disposal firm or through your local waste facility. When disposing of the device, comply with the rules and regulations that apply in your country. If in doubt, consult your local waste management facility. Proper disposal protects the environment as well as the health of your fellow human beings.

Also note that waste avoidance is a valuable contribution to environmental protection. Repairing a device or passing it on to another user is an ecologically valuable alternative to disposal.

You can return your old device to Thomann GmbH at no charge. Check the current conditions on www.thomann.de.

If your old device contains personal data, delete those data before disposing of it.

Other ManualsLib Projects

 www.manualslib.com

 www.manualslib.de

 www.manualslib.es

 www.manualslib.fr

 www.manualslib.nl

 www.manualslib.mx

 www.manualslib.tech 30+ Languages